C&I string power plant zero export solution with DataHub1000

ENGLISH

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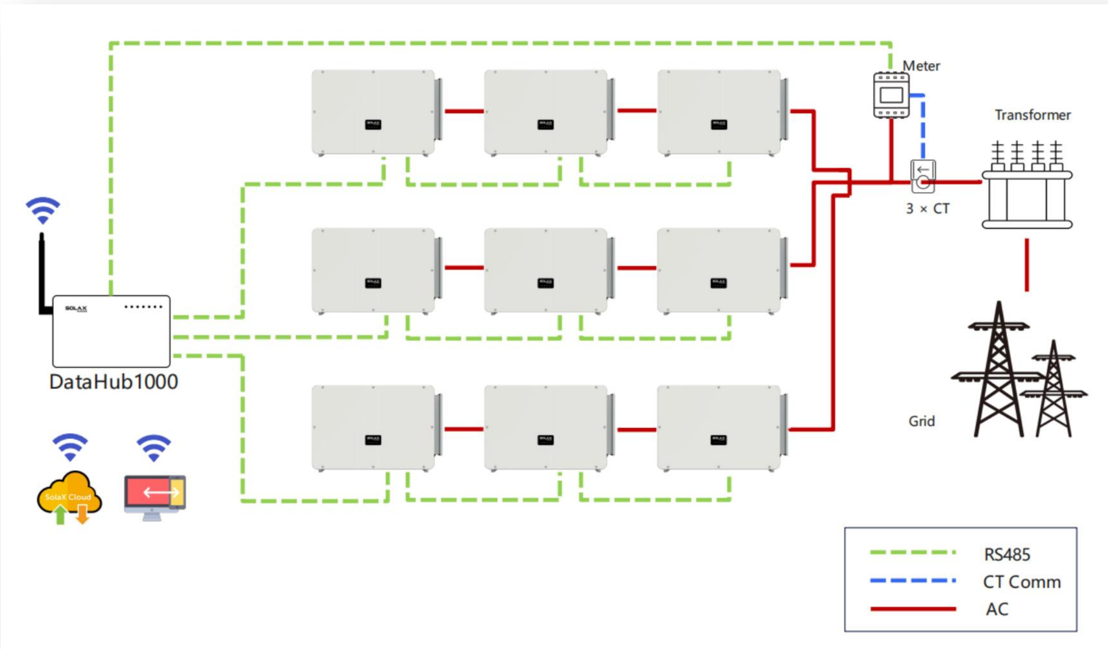
In this article, we will introduce the latest parallel solution with SolaX Power's C&I string inverters and DataHub1000 and achieve zero feed-in control.

NOTICE

- The installation shall be operated and maintained by qualified electrician or certified personnel by SolaX.
- Do not turn on the system before making sure all installations are correctly done according to the installation guide and electrical requirements from national grid.
- It is the responsibility of the site operator to set the correct feed-in power limit according to the local grid requirements. If you are not sure about the requirements, please contact the grid operator.

Parallel System Diagram with zero feed-in control

With all C&I inverters connected to the 3 available RS485 ports on DataHub1000, the output power and export power of the whole power plant can be set and controlled in accordance to the site requirements. In the meanwhile, all the inverter data can be monitored on SolaX Cloud platform.



Device List

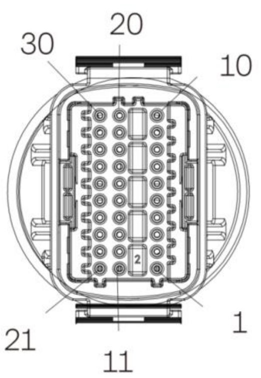
String Inverters	Datalogger	Meter	Wires
Three-phase: X3-FTH 75-150kW X3-MGA 40-60kW	DataHub1000	DTSU666-CT	PV Cables AC Cables
		CT	Communication Cables

Note: DataHub1000 has built-in Wi-Fi and LAN functions, which is ready to use out of box. Thus no additional Pocket Wi-Fi/LAN is required in this system.

For more C&I inverter solution details, please refer to the article on this link here.

System COM Connection

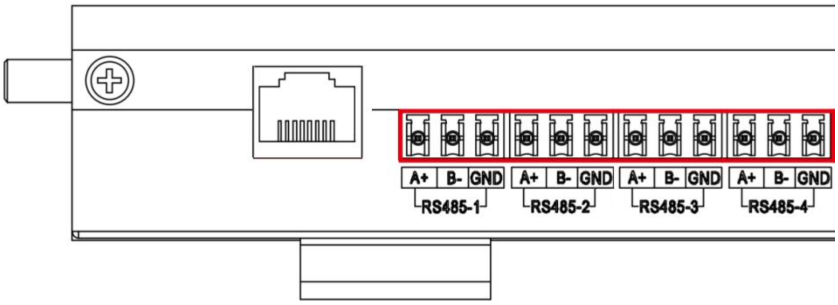
On inverter side - connect the RS485 comm cable in daisy chain to COM port at the bottom of each inverter as below:



Pin	Definition
1	RS485A IN+
2	RS485B IN-
3	RS485 IN-GND
4	RS485A OUT+
5	RS485B OUT-
6	RS485 OUT-GND

Note: From 4&5 on the 1st inverter to 1&2 on the 2nd inverter, and then 4&5 on the 2nd one to 1&2 on the next one, until 1&2 on the last one. From 4&5 on the last one to DataHub1000.

On the DataHub1000 side - connect the other end of the RS485 comm cable to the corresponding RS485 port on DataHub1000:



PIN	Definition	Function
RS485-1	RS485-1A RS485-1B GND	This port can communicate with up to 20 inverters in cascading daisy chain.
RS485-2	RS485-2A RS485-2B GND	This port can communicate with up to 20 inverters in cascading daisy chain.
RS485-3	RS485-3A RS485-3B GND	This port can communicate with up to 20 inverters in cascading daisy chain.
RS485-4	RS485-4A RS485-4B GND	This port can communicate with smart meter like DTSU666-CT from SolaX.

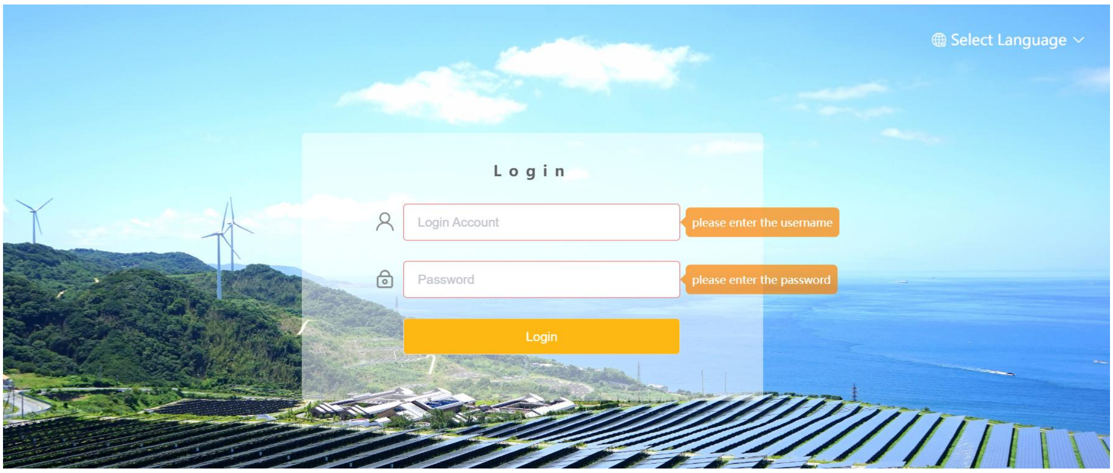
System configuration

Once all inverters are properly connected and linked to Datahub1000, we can start the system to set up the configuration.

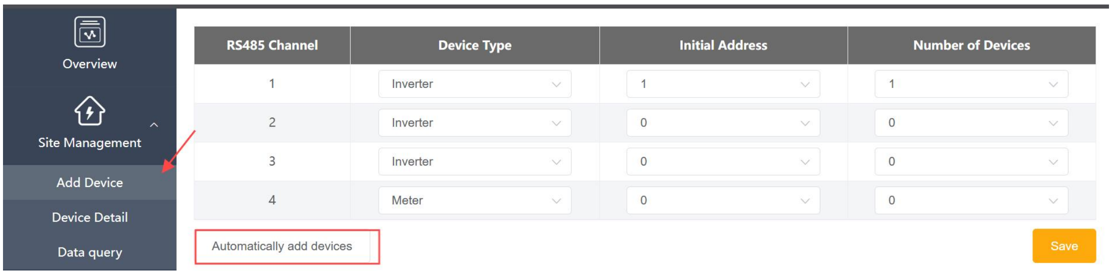
Please refer to the user manual of DataHub1000 for more details of system configuration:
https://kb.solaxpower.com/data/detail/ff80808184a409170184c14d7e6a0014

1. Login to Datahub1000 management page

After connecting a laptop to the WiFi hotspot of DataHub, please login to the configuration page via http://192.168.10.10 on browser.



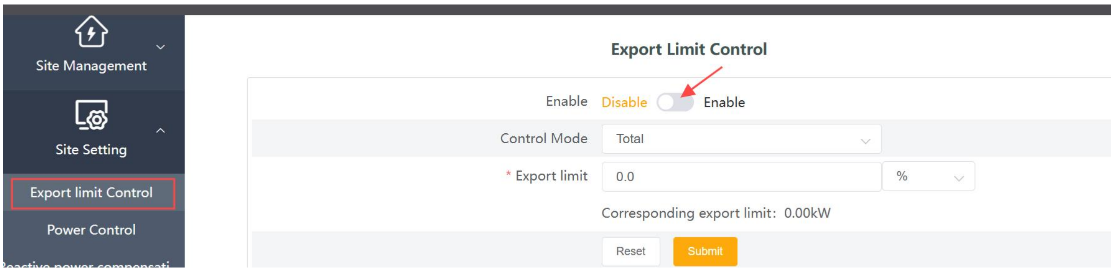
2. Add the correct inverter type and quantity by "Add device"



For Forth&Mega inverter, you click on the "Automatically add devices" function below.

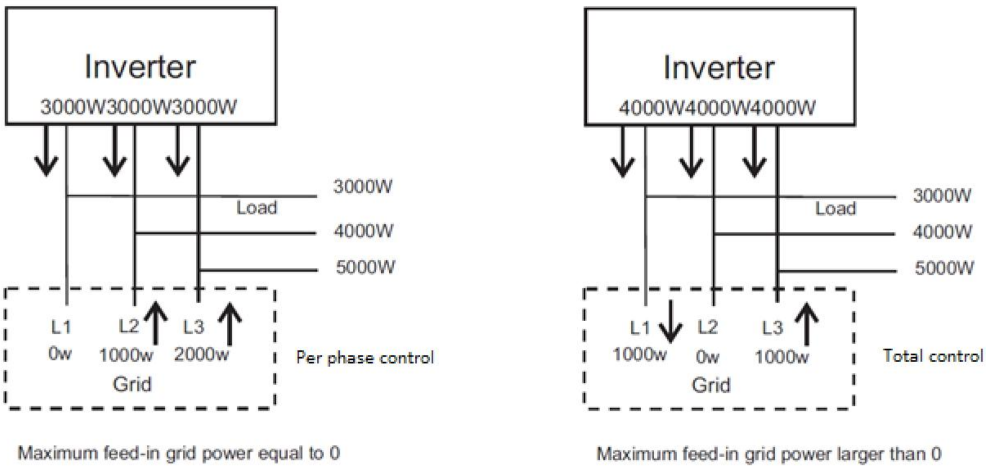
3. Set export control limit

Click "Site Setting" -> "Export limit Control"



There are two control modes for selection, **Total** and **Per Phase**.
"Total": The Site Limit is the total export power (the combined production minus the combined consumption) on all the phases combined. Reverse current on one

phase will count as negative power and can compensate for another phase.
"Per Phase": For three phase inverter connections, the inverter sets the limit on each phase to 1/3 of the total site limit. Use this mode if there is a limit on each individual phase.



Note: Per Phase mode is not applicable to three phase three wire grid.